

Class 16: Obtaining and processing SRA datasets on AWS

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##Downstream Analysis

R function called `tximport()` in the `tximport` package, which enables straightforward import of Kallisto results

```
# setup the folder and filenames to read
library(tximport)
folders <- dir(pattern="SRR21568*")
samples <- sub("_quant", "", folders)
files <- file.path( folders, "abundance.tsv" )
names(files) <- samples

txi.kallisto <- tximport(files, type = "kallisto", txOut = TRUE)
```

Note: importing ``abundance.h5`` is typically faster than ``abundance.tsv``

reading in files with `read_tsv`

1 2 3 4

```
head(txi.kallisto$counts)
```

	SRR2156848	SRR2156849	SRR2156850	SRR2156851
ENST00000539570	0	0	0.00000	0
ENST00000576455	0	0	2.62037	0
ENST00000510508	0	0	0.00000	0
ENST00000474471	0	1	1.00000	0
ENST00000381700	0	0	0.00000	0
ENST00000445946	0	0	0.00000	0

```
colSums(txi.kallisto$counts)
```

SRR2156848	SRR2156849	SRR2156850	SRR2156851
2563611	2600800	2372309	2111474

```
sum(rowSums(txi.kallisto$counts)>0)
```

```
[1] 94561
```

Before subsequent analysis, we might want to filter out those annotated transcripts with no reads:

```
to.keep <- rowSums(txi.kallisto$counts) > 0
kset.nonzero <- txi.kallisto$counts[to.keep,]
```

And those with no change over the samples:

```
keep2 <- apply(kset.nonzero,1,sd)>0
x <- kset.nonzero[keep2,]
```

##PCA

```
pca <- prcomp(t(x), scale=TRUE)
```

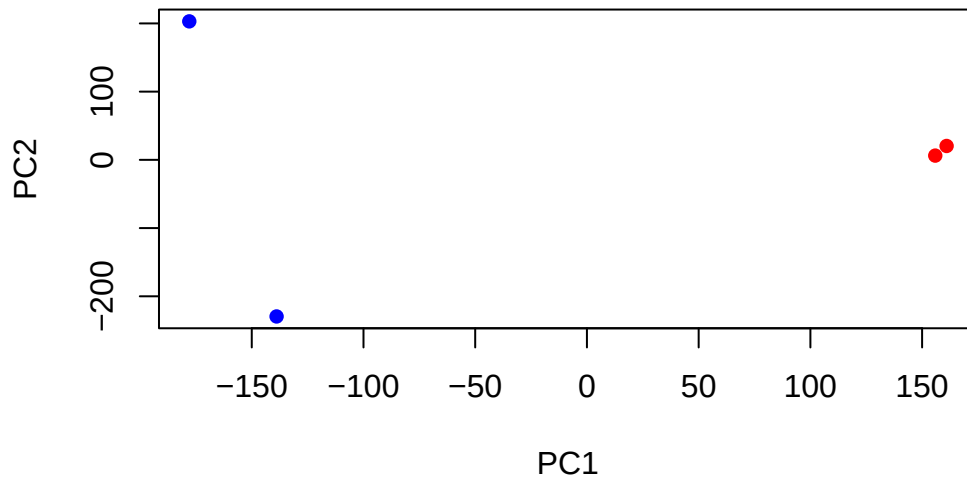
```
summary(pca)
```

Importance of components:

	PC1	PC2	PC3	PC4
Standard deviation	183.6353	177.3561	171.3005	4.91e-11
Proportion of Variance	0.3568	0.3328	0.3105	0.00e+00
Cumulative Proportion	0.3568	0.6895	1.0000	1.00e+00

Now we can use the first two principal components as a co-ordinate system for visualizing the summarized transcriptomic profiles of each sample:

```
plot(pca$x[,1], pca$x[,2],  
     col=c("blue", "blue", "red", "red"),  
     xlab="PC1", ylab="PC2", pch=16)
```

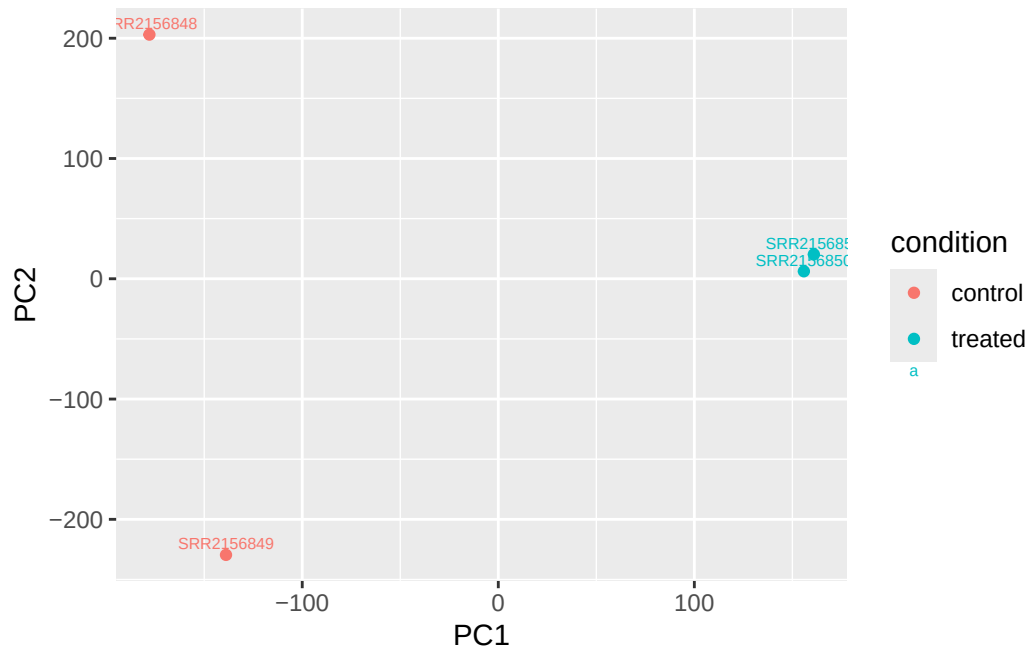


PC1 vs PC2

```
library(ggplot2)
```

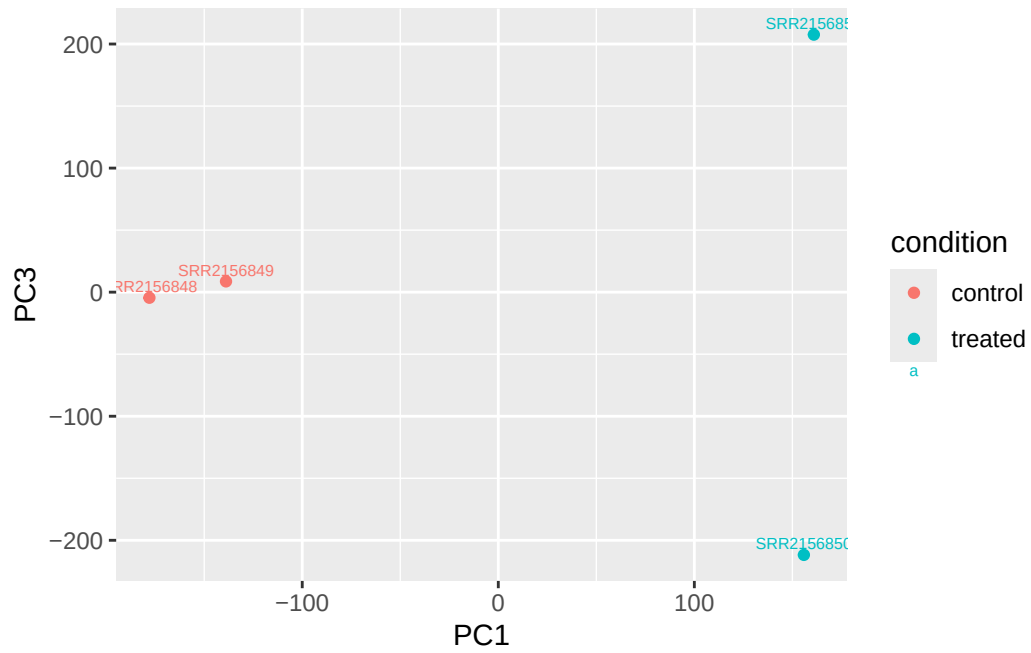
Warning: package 'ggplot2' was built under R version 4.5.2

```
condition <- c("control", "control", "treated", "treated")  
ggplot(pca$x) + aes(PC1, PC2, col=condition, label=samples)+  
  geom_point()+  
  geom_text(vjust=-0.5, size=2)
```



PC1 vs PC3

```
ggplot(pca$x) + aes(PC1, PC3, col=condition, label=samples)+
  geom_point()+
  geom_text(vjust=-0.5, size=2)
```



PC2 vs PC3

```
ggplot(pca$x) + aes(PC2, PC3, col=condition, label=samples)+
  geom_point()+
  geom_text(vjust=-0.5, size=2)
```

